Agri Doc: An Integrated Smart System for Cashew Cultivation

Project ID – 25-26J-175

Project Proposal Report

Usman M.H- IT22351104

Bachelor of Science (Hons) Degree in Information Technology
Specializing in

Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka(SlIIT)

August 2025

Agri Doc: Integrated Pest Identification System for Cashew Plants

Project ID – 25-26J-175

Project Proposal Report

Usman M.H- IT22351104

Bachelor of Science (Hons) Degree in Information Technology
Specializing in

Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka(SlIIT)

August 2025

ABSTRACT

Cashew cultivation is a vital agricultural sector in many tropical regions, but it faces significant threats from pest infestations and reduce crop yield, quality, and farmer income. Traditional pest management methods mostly depend on manual inspection and large-scale pesticide use. These approaches are not only slow and costly but also harmful to the environment and human health. Farmers often struggle to detect pests at an early stage, which leads to delayed preventive actions and greater crop losses. With the rapid development of artificial intelligence and computer vision, there is now an opportunity to create efficient, accurate, and eco-friendly solutions to support cashew farming.

This research proposes a Pest Identification System for Cashew Plants built on the YOLOv8 deep learning model. A custom dataset of cashew pests was collected from local farms, and data augmentation techniques were applied to enhance robustness under different lighting, background, and environmental conditions. The trained model demonstrates strong performance by detecting even very small pests on cashew leaves and nuts in real time, while effectively distinguishing them from complex farm backgrounds. To improve transparency and farmer trust, the system incorporates Grad-CAM visualization, which highlights the regions of each image that influenced the detection, allowing users to confirm results visually.

For practical deployment, a mobile and web application was developed that enables farmers to capture or upload photos directly from the field and instantly identify pests. The system provides a fast, accurate, and user-friendly solution, reducing dependency on pesticides and supporting sustainable pest management. By combining advanced object detection with explainable AI, this research delivers a scalable and environmentally conscious tool that empowers farmers and strengthens long-term cashew cultivation.

Keywords

Cashew Pest Detection, Deep Learning, YOLOv8, Explainable AI,smart Agriculture, Grad-CAM

Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where acknowledgement is made in the text.

Name	Student ID	Signature
Usman M.H	IT22351104	[redacted]

The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

27/8/25

....

Signature of the Supervisor Date

Ms. Shashika Lokuliyana

27/8/25

....

Signature of the Co-Supervisor Date

Ms.Suranjini Sliva

Table of Contents

ABSTRACT 3
Keywords 3
Declaration 4
List of figures 6
List of Tables 6
LIST OF ABBREVIATIONS 6
1 INTRODUCTION 7
1.1 Background and Literature Survey 8
1.2Research Gap 10
1.3 Research Problem 11
2 OBJECTIVES 12
2.1 Main Objectives 12
2.2 Specific Objectives 12
3 METHODOLOGY 13
3.1 Component System Architectur 13
3.2 Gantt chart 15
4 Project requirements 16
4.1 Functional Requirements 16
4.2 Nonfunctional Requirements 16
4.3 Technology Selection 17
4.4 Expected Test Cases 18
5 Budget Table 19
6 Work Breakdown Structure 19
7 CONCLUSIONS 20
REFERENCES 21

List of figures

Figure:4 1 Gantt chart 15

List of Tables

Table 1 List of Abbreviation 6
Table 2 compared to old model.....9

LIST OF ABBREVIATIONS

Key Words	Meaning
AI	Artificial intelligent
DB	Database
UI	User Interface
Grad-CAM	Gradient-weighted Class Activation Mapping

Table 1 List of Abbreviation

1 INTRODUCTION

One of the most critical challenges in cashew production is pest infestation, which reduces crop yield and threatens farmer livelihoods. Traditional pest management methods rely on manual inspection and heavy pesticide use, which not only demand significant labor but also cause environmental harm and degrade soil quality. With the rise of artificial intelligence (AI) and deep learning, it is now possible to develop intelligent systems that enable accurate pest detection and provide targeted, sustainable pest control strategies.

Recent advances in computer vision and object detection models have greatly improved the ability to identify agricultural pests. Among these, YOLOv8 and transformer-based models have emerged as powerful frameworks for real-time object detection, offering superior accuracy and speed, even under challenging field conditions such as variable lighting, complex cashew leaf textures, and small pest sizes. By leveraging these models, farmers can detect pests and diseases in cashew plants more reliably, enabling proactive and precise management strategies.

To achieve this, the proposed system will develop a specialized dataset of cashew pests and diseases, collected and annotated from local plantations, and enhanced with data augmentation techniques for robustness. The detection model will be trained and fine-tuned on this dataset to optimize performance in real-world scenarios. In addition to detection, the system incorporates explainable AI (Grad-CAM visualization) to highlight affected regions of the leaf or nut, increasing transparency and farmer trust in the results. Once pests or diseases are identified, the system will recommend appropriate organic or inorganic control methods, prioritizing eco-friendly approaches to safeguard beneficial insects and support sustainable farming.

The solution will be delivered through a farmer-friendly mobile application, enabling users to capture and upload images for instant analysis. The app will provide real-time detection, visual explanations, and practical treatment suggestions in local languages, ensuring accessibility to a wide farming community. By combining advanced deep learning models with sustainability-focused recommendations, this research aims to deliver a practical, scalable, and environmentally responsible solution to improve pest management and promote resilient cashew cultivation.

1.1 Background and Literature Survey

Literature Review

Precision agriculture increasingly relies on artificial intelligence (AI) and computer vision for automated pest detection, which is critical for maintaining crop yield and reducing losses. Traditional methods of pest identification involve manual inspection, which is labor-intensive, time-consuming, and prone to human error. To overcome these limitations, deep learning models, particularly convolutional neural networks (CNNs), have been applied for automated detection and classification of pests in various crops [3], [5]. CNN-based methods provided an initial framework for recognizing pest patterns; however, their real-time performance and ability to detect very small pests were limited.

The YOLO (You Only Look Once) series of object detection models revolutionized pest detection in agriculture by combining accuracy with real-time detection capability. Early versions, such as YOLOv3 and YOLOv5, were applied for general crop pest identification and achieved notable success in detecting medium-to-large pests [11], [7]. However, these models demonstrated significant limitations in detecting extremely small pests, specifically those smaller than 2 pixels in high-resolution images. Small-object detection is a crucial requirement in cashew plantations, where many pests, such as the tea mosquito bug (Helopeltis spp.) or cashew nut borer (Toxoptera spp.), appear very small but cause substantial crop damage [1], [6].

CNN-based and YOLOv3/v5 models also faced challenges in field conditions due to complex backgrounds, lighting variations, and overlapping leaves or nuts. Such environmental factors often resulted in false negatives or misclassification, limiting practical applicability [3], [4]. Furthermore, earlier models primarily focused on offline detection, where images needed to be captured and processed later, delaying real-time intervention. These constraints highlighted the need for more advanced models capable of detecting tiny pests accurately and providing live, actionable insights for farmers.

YOLOv8, the latest evolution in the YOLO series, addresses these limitations by integrating advanced feature extraction, transformer-based attention mechanisms, and optimized small-object detection capabilities [11], [8]. Compared to YOLOv5 and CNN-based methods, YOLOv8 demonstrates superior accuracy for pests smaller than 2 pixels, thanks to its enhanced multi-scale feature aggregation and refined detection heads. Moreover, YOLOv8 enables live detection, allowing real-time identification of pests directly in the field. This capability can be integrated with GPS or location-tagging tools, providing precise pest location mapping, which is crucial for site-specific interventions in cashew plantations.

Recent studies indicate that YOLOv8 not only improves detection accuracy but also reduces computational cost, enabling deployment on mobile or edge devices [11], [2].

This advancement is particularly relevant for small-scale farmers in remote areas, where access to high-performance computing resources is limited. Additionally, YOLOv8's compatibility with explainable AI methods such as Grad-CAM facilitates interpretability, allowing users to understand which regions of an image triggered the detection, enhancing trust and adoption [7], [6].

Despite these advancements, the majority of prior research on cashew pest detection has relied on general datasets or models not specifically optimized for cashew pests. Liu et al. (2024) applied EfficientNetv2-S for cashew nut pest classification, but the system was not designed for live field deployment [1]. Similarly, Zhang et al. (2023) used YOLOv5 for pest detection in general crops, which could not reliably detect sub-2-pixel pests [2]. Therefore, implementing YOLOv8 in cashew plantations represents a significant step forward in bridging this gap, offering a solution that combines small-object detection, live monitoring, and location-aware precision for effective pest management.

COMPARISON TO OLD MODELS

Old Models (CNN, YOLOv3, YOLOv5)	New Model (YOLOv8)
CNN Offline processing, poor detection of tiny pests (< 2 pixels), sensitive to lighting, occlusion, and complex field backgrounds. Computationally heavy, not real-time, requires large datasets.	Live detection in the field, accurate for sub-2-pixel pests. Robust under complex conditions. Lightweight and deployable on mobile/edge devices. Supports GPS/location tagging and explainable AI.
YOLOv3 Limited real-time detection, struggles with small pests and overlapping objects. Accuracy drops in field conditions. Tested on general crop datasets.	Transformer-based enhancements, multi-scale feature aggregation. Cashew-specific training possible. High accuracy for tiny pests and real-time monitoring.

Table 2 compared to old model

1.2Research Gap

Despite significant advancements in automated pest detection using deep learning, previous studies on cashew pest management have predominantly relied on traditional CNN models and earlier YOLO versions, such as YOLOv3 and YOLOv5 [1]–[5]. These approaches are limited in their ability to detect extremely small pests, particularly those smaller than 2 pixels, which are common in cashew plantations. Tiny pests, such as *Helopeltis spp.* and *Toxoptera spp.*, can cause substantial crop damage, yet existing models often fail to identify them accurately, leading to delayed or ineffective interventions.

Moreover, earlier models mostly operate in offline or near-real-time settings, which prevents immediate pest management actions in the field. The detection accuracy of these models is also compromised under complex agricultural conditions, including overlapping leaves, variable lighting, and background noise, resulting in false negatives and misclassifications. Additionally, prior research rarely incorporates location-aware detection, limiting the capacity for site-specific pest control strategies.

Another limitation is the computational demand of traditional models, which restricts their deployment on mobile or edge devices commonly used in remote plantations. Furthermore, explainability in pest detection has been largely overlooked, making it difficult for farmers and agronomists to interpret model decisions and trust automated recommendations.

Thus, there exists a clear research gap for a cashew-specific pest detection system that can provide live, real-time monitoring, accurately detect tiny pests under complex field conditions, support GPS-based location mapping, and be deployable on mobile/edge devices with interpretable outputs. The latest YOLOv8 model, with its transformer-based enhancements, multi-scale feature aggregation, and lightweight design, offers the potential to address these gaps, yet its application for cashew pest detection remains underexplored.

1.3 Research Problem

The main problem in cashew farming is poor pest and disease management, which can reduce crop yield, increase costs, and harm the environment because of too much pesticide use. Traditional methods rely on farmers checking the plants by hand or spraying pesticides without clear guidance. This takes a lot of time and effort and is often not accurate, leading to late detection or wrong treatment.

There is no cashew-specific AI system that can quickly identify pests and diseases and give farmers useful advice. Most existing AI models are made for general crops or lab conditions and do not work well for cashew farms. They struggle with small pests, complex leaf patterns, and real field conditions.

This study will answer these questions:

1. How can YOLOv8 and transformer models detect and identify cashew pests and diseases accurately in real-time?
2. How can Grad-CAM show farmers exactly where the pest is, so they trust the system?
3. How can a mobile app give farmers simple and eco-friendly advice on pest control, reducing pesticide use and protecting the environment?

The goal is to build a smart cashew pest control system that helps farmers spot pests quickly, take the right actions, and grow cashew crops in a safe and sustainable way.

2 OBJECTIVES

2.1 Main Objectives

The main objective of this project is to develop a Smart Cashew Pest Control System that uses AI and deep learning to detect pests and diseases in cashew plants and provide farmers with real-time, practical advice. Traditional pest management relies on manual checks and spraying pesticides without guidance, which is slow, inaccurate, and can harm the environment. This system will use YOLOv8 with transformer models to accurately identify pests, and Grad-CAMto show exactly where the problem is, helping farmers trust the results. It will give eco-friendly treatment recommendations, prioritizing organic methods before chemical pesticides, and a mobile app will allow farmers to take photos, get instant detection results, and follow easy guidance. The goal is to reduce crop losses, minimize chemical use, and make cashew farming more efficient and sustainable.

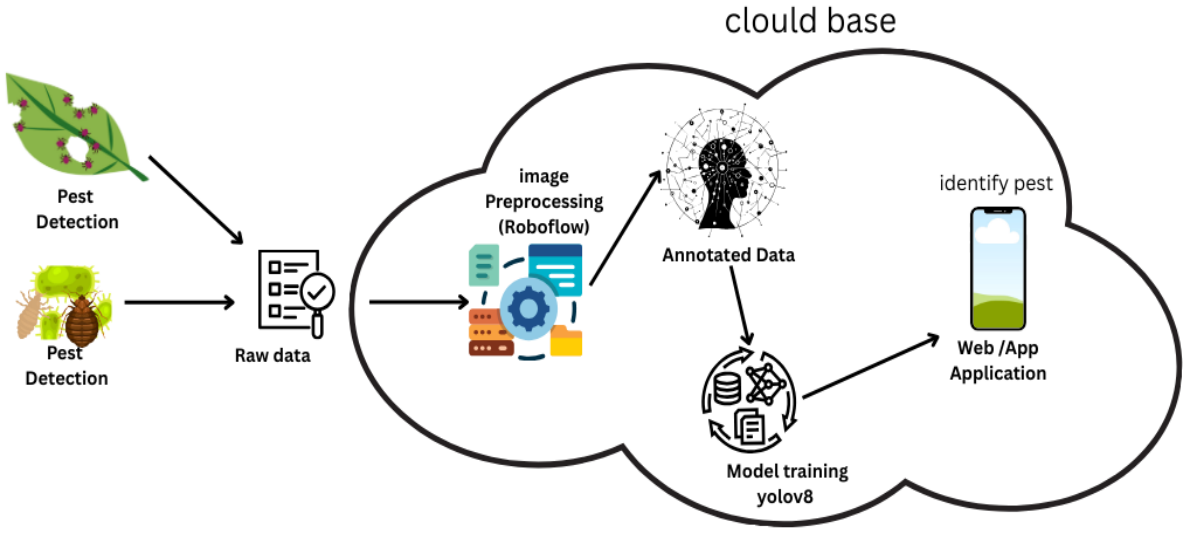
2.2 Specific Objectives

1. To develop a YOLOv8-based detection system enhanced with transformer models that can accurately identify cashew pests and diseases in real-time, even when pests are very small or hidden under complex leaf patterns.
2. To develop a Grad-CAM–based visualization module that highlights the exact location of pests in cashew plant images, improving interpretability and increasing farmers' trust in the system.
3. To develop a mobile application that allows farmers to capture images of cashew plants, receive instant detection results, and access simple, eco-friendly pest control recommendations—prioritizing organic methods before chemical treatments to ensure environmental sustainability.

3 METHODOLOGY

The development of the Smart Cashew Pest Control System follows a clear methodology to ensure accurate pest detection and practical recommendations for farmers. The process starts with data collection, where images of cashew leaves, nuts, and shells will be captured from plantations to build a specialized pest and disease dataset [1]. This dataset will be enhanced with data augmentation to make the model more robust in different lighting, backgrounds, and field conditions [2].

3.1 Component System Architectu



Finger 3.1

Finger3.1 get the data collection, after which images undergo preprocessing (e.g., via Roboflow). A deep learning model (YOLOv8) is then trained for pest detection. The trained model is deployed in a web or mobile application to perform real-time pest identification, completing an end-to-end system from data to actionable results.

Next, the collected data undergoes preprocessing, which includes removing low-quality images, resizing to a uniform dimension, and annotating pest and disease regions using bounding boxes [3]. The cleaned dataset is then used to train a hybrid deep learning model combining YOLOv8 for real-time object detection and a transformer-based module for improved feature extraction and classification [4]. To enhance interpretability, Grad-CAM is applied to visualize the regions of interest within the plant images, helping farmers understand the model's decision-making process [5].

The system is designed to provide eco-friendly pest control recommendations, prioritizing organic and biological methods before suggesting chemical interventions to minimize environmental impact [6]. A user-friendly mobile and web application is developed to allow farmers to upload images, receive real-time detection results, and access practical guidance in local languages [7]. The application also includes features such as dashboards, notifications, and feedback mechanisms to support continuous improvement and user engagement [8].

By combining AI-based detection, explainable visualizations, and mobile access, this system aims to reduce crop losses, minimize chemical use, and support sustainable cashew farming [9]. Farmers will be empowered to make quick, informed decisions to manage pests effectively and protect their crops [10].

3.2 Gantt chart

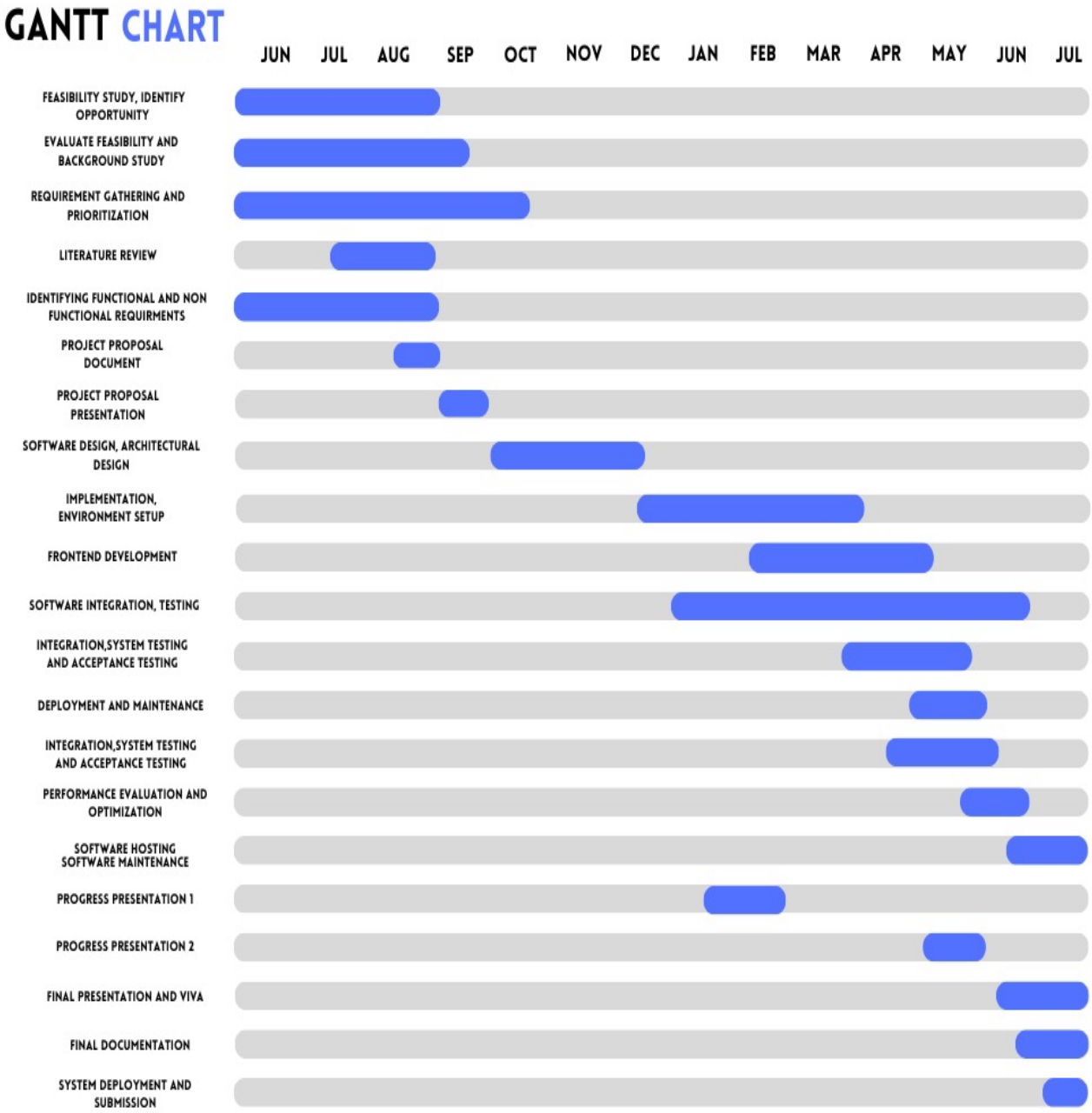


Figure:3.2 Gantt char

4 Project requirements

4.1 Functional Requirements

1. The system must accurately detect pests and diseases on cashew leaves, nuts, and shells from images and performed in real-time and provide results.
2. The system must classify the type of pest display bounding boxes around detected pests/diseases for clear localization.
3. The system must accept images with a minimum resolution of 640×640 pixels and For small pests024×1024 to ensure sufficient detail for detection
4. The system Ensure good lighting (ideally >300 lux) and minimal shadows.
5. The system must show visual explanations (using Grad-CAM) to highlight where the pest or disease is located.

4.2 Nonfunctional Requirements

1. Performance
   - The system must process and analyze a captured image within 5 seconds to ensure real-time usability.

2. Accuracy
   - The system must maintain a minimum of 90% detection accuracy for common cashew pests and diseases under recommended image conditions.

3. Reliability and Availability

   - The system must be available 99% of the time with minimal downtime for maintenance.

   - The system must ensure robust error handling for invalid image uploads or low-quality images.

4. Scalability
   - the system must allow future expansion to support additional pest types, diseases, and larger datasets.

5. Usability
    - The system must provide a simple and intuitive mobile interface for image upload and result visualization.

6. Portability
    - The system must support deployment on cloud servers to allow remote access and scalability.

4.3 Technology Selection

1. Programming Language
    - Python→ Used to build machine learning models, process images, and handle the backend logic.

2. Machine Learning and Deep Learning

    - Yolov8 → using data collection to train the mode
    - Google colab → for using model training for the dataset

3. Computer Vision

    - OpenCV → To process images, detect pests, and extract features.
    - Pillow → For basic image editing like resizing, cropping, or adjusting colors.

4. Dataset Management

    - Roboflow→allow images to be labeled and annotated using bounding boxes to identify pests and affected areas

5. Backend Frameworks
    - Flask→ A lightweight Python framework to create the backend, allowing farmers to upload images and get pest detection results.

6. Frontend Frameworks
    - HTML/CSS/JavaScript
    - React

7. Designing Tools
    - UI design: Figma
    - Diagram designs: Draw.io, Canva

8. Deployment and Hosting
    - AWS/Google Cloud/Azure

9.Version Control → Git/GitHub

4.4 Expected Test Cases

1. **Test Case 1**
   An image of a healthy, pest-free cashew leaf is uploaded. The system should correctly identify it as healthy and indicate that no pests are detected.

2. **Test Case 2**
   An image of a cashew leaf with a clearly visible pest (e.g., Tea Mosquito Bug) is uploaded under proper lighting conditions (>300 lux). The system should correctly identify the pest, draw a bounding box around it, and provide a specific treatment recommendation.

3. **Test Case 3**
   An image of a cashew leaf with a very small or partially hidden pest is uploaded. The system should successfully detect the pest and accurately classify it, demonstrating its ability to identify challenging infestations.

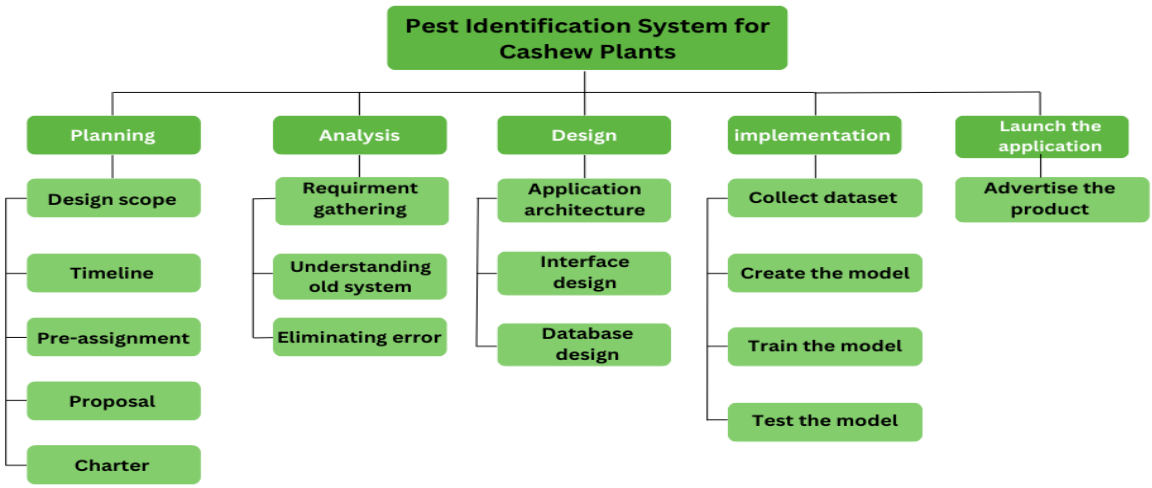
4. **Test Case 4**
   A blurry or low-quality image of a cashew plant is uploaded. The system should handle the error gracefully by not crashing and should display a user-friendly message prompting for a better image.

5. **Test Case 5**
   A user uploads a batch of five images containing a mix of healthy and infected leaves. The system should process all images successfully and display the correct detection results for each individual image in the user interface.

5 Budget Table

Item	Description	Cost Estimation (LKR)
Deploy in play store	12-month hosting plan	32000
Google Cloud	12-month	9100
Google colab	For Model train dataset	15100
Total		56200

6 Work Breakdown Structure



7 CONCLUSIONS

This project introduces an innovative Cashew Pest Control System that leverages AI, computer vision, and data-driven decision-making to improve pest management practices. By enabling farmers to detect pests and diseases in real time, the system provides accurate and timely recommendations for targeted pest control measures. This ensures that farmers can take immediate action, reducing potential crop damage.

Through AI-based detection and the integration of farm data, the system helps prevent major crop losses while reducing excessive pesticide use. By focusing only on affected areas, the system not only minimizes costs but also protects beneficial insects and supports environmentally friendly farming practices.

The inclusion of remote monitoring and a user-friendly mobile interface further enhances the practicality of the system. Farmers are empowered to manage pest control operations efficiently, saving both time and labor. This accessibility makes the system easy to adopt, even for farmers with limited technical knowledge.

Another key strength of this project is its affordability and scalability. The solution can be applied to both small-scale farms and large cashew plantations, making it suitable for a wide range of users. Future enhancements may include AI-driven disease prediction, yield forecasting, and automated decision support systems to further improve agricultural productivity.

In conclusion, this project presents a smart, sustainable, and technology-driven solution for cashew pest management. By improving crop quality, increasing yield, and promoting sustainable farming, it provides long-term benefits to farmers and contributes to the advancement of modern agriculture.

REFERENCES

[1] R. Liu, H. Yang, X. Zhang, and J. Yang, “Classification of cashew nut pests and diseases based on EfficientNetv2-S,” in *2024 IEEE International Conference on Machine Vision, Pattern Recognition (MVIPIT)*, 2024, pp. 140–144, doi: 10.1109/MVIPIT65697.2024.00032.

[2] L. Zhang, C. Zhao, Y. Feng, and D. Li, “Pests identification of IP102 by YOLOv5 embedded with the novel lightweight module,” *Agronomy*, vol. 13, no. 6, p. 1583, 2023, doi: 10.3390/agronomy13061583.

[3] T. Wu, G. Wu, and J. Tao, “A research review of pest identification and detection based on deep learning,” in *2022 34th Chinese Control and Decision Conference (CCDC)*, 2022, pp. 6076–6080.

[4] H. Kuzuhara, H. Takimoto, Y. Sato, and A. Kanagawa, “Insect pest detection and identification method based on deep learning for realizing a pest control system,” in *2020 59th Annual Conference of the Society of Instrument and Control Engineers of Japan (SICE)*, 2020, pp. 709–714, doi: 10.23919/SICE.2020.9240332.

[5] S. A. Kumar, F. Shaik, B. Yashwitha, P. Vidyavathi, A. U. Maheswari, and P. Spurthi, “Intelligence pest detection and control in agriculture using computer vision and deep learning,” in *2025 International Conference on Intelligent Computing and Control Systems (ICICCS)*, 2025.

[6] Y. Wei and X. Pan, “Improvement algorithm analysis and implementation of plant disease and pest recognition based on YOLO,” in *2024 5th International Conference on Computer Vision, Image and Deep Learning (CVIDL)*, 2024, pp. 442–446.

[7] S. K. Sundravadiyelu, K. Karthigadevi, G. E. Rani, V. S. Kumar, and M. Sakthimohan, “Enhanced YOLOv5 based approach for effective agricultural pest detection,” in *2024 Global Conference on Communications and Information Technologies (GCCIT)*, Bangalore, India, Oct. 25–26, 2024.

[8] R. Karthik, A. Ajay, A. S. Bishi, T. Illakiya, and K. Suganthi, “A deep learning approach for crop disease and pest classification using Swin transformer and dual-attention multi-scale fusion network,” *IEEE Access*, vol. 12, pp. 152639–152655, 2024, doi: 10.1109/ACCESS.2024.3481675.

[9] J. Li, Q. Zhang, H. Su, and M. Liao, “Pest and disease innovation detection model based on YOLOv5-CB algorithm,” in *Proceedings of the 43rd Chinese Control Conference*, Kunming, China, Jul. 28–31, 2024, pp. 7786–7790.

[10] C. Li, S. Chen, Y. Ma, M. Song, X. Tian, and H. Cui, "Wheat Pest Identification Based on Deep Learning Techniques," in *2024 the 7th International Conference on Big Data and Artificial Intelligence*, 2024.

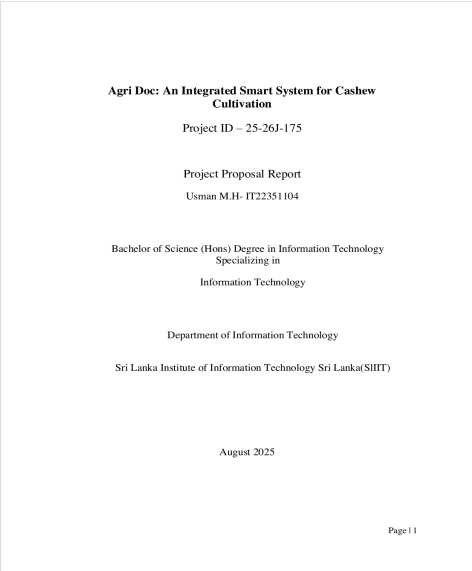
[11]A. Ali, C. Benevenuti, A. R. Al-Ajmi, and A. Kallel, “Enhancing Precision Agriculture Pest Control: A Generalized Deep Learning Approach With YOLOv8-Based Insect Detection,” *IEEE Access*, vol. 12, pp. 73248–73259, 2024, doi: 10.1109/ACCESS.2024.3417296.

Digital Receipt

This receipt acknowledges that Turnitin received your paper. Below you will find the receipt information regarding your submission.

The first page of your submissions is displayed below.

Submission author:	Kajanthan Kirubakaran
Assignment title:	Research Paper Checking
Submission title:	IT22351104_Usman.M.H_propasal report.pdf
File name:	IT22351104_Usman.M.H_propasal_report.pdf
File size:	693.57K
Page count:	22
Word count:	3,905
Character count:	25,293
Submission date:	31-Aug-2025 10:38PM (UTC+0530)
Submission ID:	2737771968



Agri Doc: An Integrated Smart System for Cashew Cultivation

Project ID – 25-26J-175

Project Proposal Report

Usman M.H- IT22351104

Bachelor of Science (Hons) Degree in Information Technology
Specializing in

Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka(SIIIT)

August 2025

Page | 1

Copyright 2025 Turnitin. All rights reserved.